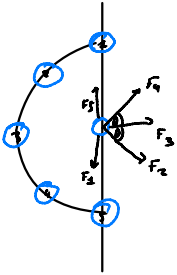


Tema 5

Preguntas T21 Tipler

Ley de Coulomb

Problema 35:



$$\sum F_y = 0$$

F_1 se cancela con F_4 y F_{4y} con F_{2y}

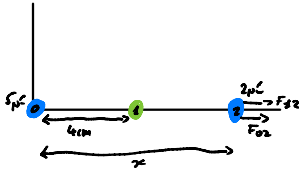
$$x: \sum F_x = F_T \quad \theta = 45^\circ$$

$$F_T = F_{2x} + F_3 + F_{4x}$$

$$F_T = \frac{kQq}{R^2} \cos(\theta) + \frac{kQq}{R^2} + \frac{kQq}{R^2} \cos(\theta)$$

$$F_T = \frac{kQq}{R^2} \left(\frac{\sqrt{2}}{2} + 1 + \frac{\sqrt{2}}{2} \right) \quad F_T = k \frac{Qq}{R^2} (1 + \sqrt{2})$$

Problema 34



$$S: x = 8 \text{ cm}, F_1 = 19,7 \text{ N}$$

$$Si: x = 17,75 \text{ cm}, F_2 = 0 \text{ N}$$

$$\text{Caso 2: } x = 17,75 \text{ cm}$$

$$F_2 = F_{02} + F_{12} = 0$$

$$F_{02} = -F_{12} \rightarrow \frac{k \cdot 5 \cdot 10^{-9} \cdot 2 \cdot 10^{-6}}{(17,75 \cdot 10^{-2})^2} = -\frac{k \cdot q \cdot 2 \cdot 10^{-6}}{(17,75 \cdot 10^{-2})^2} \rightarrow -q = 3,0 \cdot 10^{-6}$$

$$q = -3 \cdot 10^{-6} \text{ C}$$

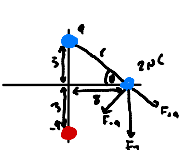
$$\text{Caso 1: } x = 8 \text{ cm}$$

$$F_2 = F_{02} + F_{12}$$

$$-19,7 = \frac{k \cdot 5 \cdot 10^{-9} \cdot 2 \cdot 10^{-6}}{(8 \cdot 10^{-2})^2} + \frac{k \cdot 2 \cdot 10^{-9} \cdot q}{(4 \cdot 10^{-2})^2} = k \cdot 2 \cdot 10^{-9} \cdot \left(\frac{5 \cdot 10^{-9}}{(8 \cdot 10^{-2})^2} + \frac{q}{(4 \cdot 10^{-2})^2} \right) \rightarrow -1,094 = \frac{5 \cdot 10^{-9}}{(8 \cdot 10^{-2})^2} + \frac{q}{(4 \cdot 10^{-2})^2}$$

$$q = -3 \mu\text{C}$$

Problema 31



$$r = \sqrt{8^2 + 3^2} = \sqrt{73} \text{ cm}$$

$$\cos(\theta) = \frac{8}{\sqrt{73}}$$

$$\sin(\theta) = \frac{3}{\sqrt{73}}$$

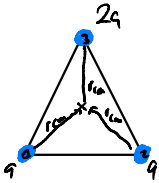
$$\sum F_x = 0 \rightarrow \text{se cancelan}$$

$$\sum F_y = 2F_{1y}$$

$$F_T = -2 \cdot \frac{k \cdot (5 \cdot 10^{-9}) (2 \cdot 10^{-6})}{73} \cdot \sin(\theta) \rightarrow F_T = -8,66 \text{ N}$$

Campo Eléctrico

Problema 47



$$x = \left(\frac{L}{2}, \frac{h}{3} \right) \quad \Rightarrow h = \sqrt{L^2 - \frac{L^2}{4}} = \frac{\sqrt{3}}{2} L$$

$$x = \left(\frac{L}{3}, \frac{\sqrt{3}}{6} L \right)$$

$$r_{\text{com}} = \frac{2}{3} h = \frac{\sqrt{3}}{3} L$$

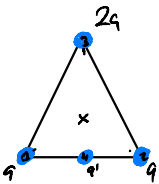
$$\sum E_x = 0 \rightarrow E_x = 0 \quad \text{entonces sus posición de } x = L/$$

$$\sum E_y = \frac{k \cdot q}{L^2/3} \cdot \sin(30) + \frac{k \cdot q}{L^2/3} \cdot \sin(30) - \frac{k \cdot 2q}{L^2/3}$$

$$\sum E_y = \frac{k \cdot q}{L^2} \left(\frac{1}{1/3} \cdot \frac{1}{2} \cdot L - \frac{2 \cdot 1/3}{1/3} \right) \rightarrow E_y = -3 \frac{kq}{L^2} \quad E_u = \frac{kq}{d^2} = 3 \frac{kq}{L^2} \rightarrow \frac{1}{d^2} = \frac{3}{L^2}$$

$$d = \frac{L}{\sqrt{3}} \rightarrow \text{Pos } q_u = \left(\frac{L}{2}, \frac{\sqrt{3}}{3} L \right)$$

Problema 48



$$x = \left(\frac{L}{2}, \frac{h}{3} \right) \quad \Rightarrow h = \sqrt{L^2 - \frac{L^2}{4}} = \frac{\sqrt{3}}{2} L$$

$$x = \left(\frac{L}{3}, \frac{\sqrt{3}}{6} L \right)$$

$$r_{\text{com}} = \frac{2}{3} h = \frac{\sqrt{3}}{3} L$$

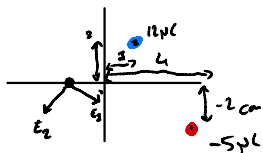
$$\sum E_x = 0 \rightarrow E_{1x} - E_{2x} = 0 \rightarrow \text{se cumple siempre}$$

$$\sum E_y = 0$$

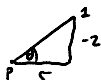
$$\frac{k \cdot q \cdot \sin(30)}{L^2/3} + \frac{k \cdot q \cdot \sin(30)}{L^2/3} - \frac{k \cdot 2q}{L^2/3} = 0$$

$$\Rightarrow \frac{k \cdot q}{L^2} \left(\frac{3}{2} + \frac{3}{2} - 6 \right) + \frac{12 k \cdot q'}{L^2} = 0 \rightarrow \frac{3 k q}{L^2} = \frac{12 k q'}{L^2} \rightarrow q' = \frac{3}{12} q = \frac{q}{4}$$

Problema 43



Para 1:



$$r^2 = 5^2 + (-2)^2$$

$$r = \sqrt{29}$$

$$\cos(\theta) = \frac{5}{\sqrt{29}}$$

$$\sin(\theta) = -\frac{2}{\sqrt{29}}$$

Para 2:



$$r = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$\cos(\theta) = \frac{2}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

$$\sin(\theta) = \frac{1}{\sqrt{5}} = \frac{\sqrt{5}}{5}$$

$$E_p = \sum E_i$$

$$E_p = E_3 + E_2 = -8219,1 \vec{i} - 14895,9 \vec{j} \text{ N/C}$$

$$E_3 : |E_1| \cos(\theta) \vec{i} + |E_1| \sin(\theta) \vec{j}$$

$$= \frac{4 \cdot 10^9 \cdot 5 \cdot 10^{-6}}{29} \cdot \frac{5}{\sqrt{29}} \vec{i} - \frac{4 \cdot 10^9 \cdot 5 \cdot 10^{-6}}{29} \cdot \frac{2}{\sqrt{29}} = 1440,7 \vec{i} - 576,3 \vec{j}$$

$$E_2 : |E_2| \cos(\theta) \vec{i} + |E_2| \sin(\theta) \vec{j} = \frac{9 \cdot 10^9 \cdot 12 \cdot 10^{-6}}{5} \left(-\frac{\sqrt{5}}{5} \vec{i} - \frac{2\sqrt{5}}{5} \vec{j} \right)$$

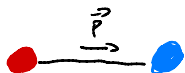
$$E_2 = -9659,81 \vec{i} - 19319,6 \vec{j} \text{ N/C}$$

Dipolos

Problema 59

$$a) \vec{p} = q \cdot d = 2 \cdot 10^{-12} \cdot 4 \cdot 10^{-3} = 8 \cdot 10^{-15} \text{ C} \cdot \text{m}$$

b)



Problema 60

$$E = 4 \cdot 10^6 \text{ V/C} \quad \vec{p} = 0,5 \cdot 10^{-9} \text{ C} \cdot \text{m}$$

$$a) \text{ Si } \vec{p} \parallel \vec{E} \rightarrow p = 0,5 \cdot 10^{-9} \text{ C} \cdot \text{m} \rightarrow \text{Equilibrio estable}$$

$$b) \text{ Si } \vec{p} \perp \vec{E} \rightarrow p = 0,5 \cdot 10^{-9} \text{ C} \cdot \text{m}$$

$$c) p = 0,5 \cdot 10^{-9} \cdot \sin(30) = 0,25 \cdot 10^{-9} \text{ C} \cdot \text{m}$$

$$d) U = -p \cdot E$$

$$\text{Si } \parallel \rightarrow U = 0 \text{ J}$$

$$\text{Si } \perp \rightarrow U = 0,5 \cdot 10^{-9} \cdot 4 \cdot 10^6 = 2 \cdot 10^{-3} \text{ J}$$

$$\text{Si } 30^\circ \rightarrow U = 1 \cdot 10^{-3} \text{ J}$$